RFID Wisp

User Manual



Read and write the filament tags of the QIDI Box, manage spools with Spoolman and print QR code labels – for Windows, Linux and Android.

Applies to RFID Wisp version 0.8.8 (as of October 2026)

> **Note:** This English manual is an automated translation of the German original. Mistakes cannot be ruled out. If something is unclear, please compare it with the German manual.

Table of Contents

Table of Contents 2

1 Introduction 4

1.1 What is RFID Wisp for? 4

1.2 Key terms 4

1.3 With and without Spoolman 5

1.4 Conventions in this manual 5

2 Requirements and Installation 6

2.1 What you need 6

2.2 Windows 6

2.3 Linux 7

2.4 Android 8

2.5 Setting up the reader (Windows and Linux) 8

3 First Start and Settings 9

3.1 Starting the app 9

3.2 The settings dialog 9

3.3 “General” tab 10

3.4 “Reader” tab (Windows and Linux only) 10

3.5 “Printers” tab 12

3.6 “Spoolman” tab 13

3.7 Exporting and importing settings 13

4 The Main Window 15

4.1 Layout 15

4.2 The menu 15

4.3 The status bar 15

4.4 Busy pointer and tooltips 16

4.5 What works when – overview 16

5 The “QIDI Data” Frame 17

5.1 Choosing printer and box 17

5.2 The slots 18

5.3 Warning “The printer has no connection to Spoolman” 19

6 The “RFID Tag” Frame 20

6.1 Layout of the frame 20

6.2 Where do the selection lists come from? 20

6.3 Reading a tag 20
6.4 Writing a tag with Spoolman 21
6.5 Writing a tag without Spoolman 23
6.6 With and without Spoolman compared 24
6.7 Special cases and errors when writing 25
6.8 Writing several tags in a row 26
7 The “QR Code” Frame 27
7.1 When does the code appear? 27
7.2 What is in the code? 27
7.3 Exporting and printing 28
8 Typical Workflows Step by Step 29
8.1 First-time setup 29
8.2 Registering new filament from another vendor with Spoolman 29
8.3 Registering new filament without Spoolman 29
8.4 Finding out what is on a tag 29
8.5 Replacing the tag of an existing spool 29
8.6 Creating a QR label for a spool in the box 30
8.7 Checking which spool is where 30
9 Using the App on Android 31
9.1 Setting up NFC 32
10 Updates 33
11 Optional: the Klipper Module rfid_bridge 34
11.1 Installation 34
11.2 How it works 35
11.3 Optional settings 36
12 Troubleshooting 38
12.1 Problems and solutions 38
12.2 The spool number is above 999 39
12.3 The log (Windows only) 39
Appendix A Layout of the Tag Data 40
Appendix B Status Bar Messages 41
Appendix C Glossary 42

1 Introduction

1.1 What is RFID Wisp for?

The filament boxes from QIDI (the **QIDI Box** on the **QIDI Plus4**, **QIDI Max4** and **QIDI Q2** printers) recognize the loaded filament by a small RFID chip on the spool. It is a **MIFARE Classic 1K** tag. It holds the **filament type (material)**, the **vendor**, the **color** and – if you set it up that way – a **spool number** that identifies the spool in the filament manager **Spoolman**.

With RFID Wisp you can:

- **see what the box has recognized**: material, vendor, color, spool number and remaining weight for each of the four slots of a QIDI box (the "QIDI Data" frame),
- **read RFID tags** and show what is on them,
- **write RFID tags**, for example for filament from other vendors that comes without a suitable chip – with or without Spoolman,
- **create spools in Spoolman** and link them to the tag right away,
- print a **QR code label** of the spool or save it as a PDF,
- use the app in **German, English or Ukrainian**.

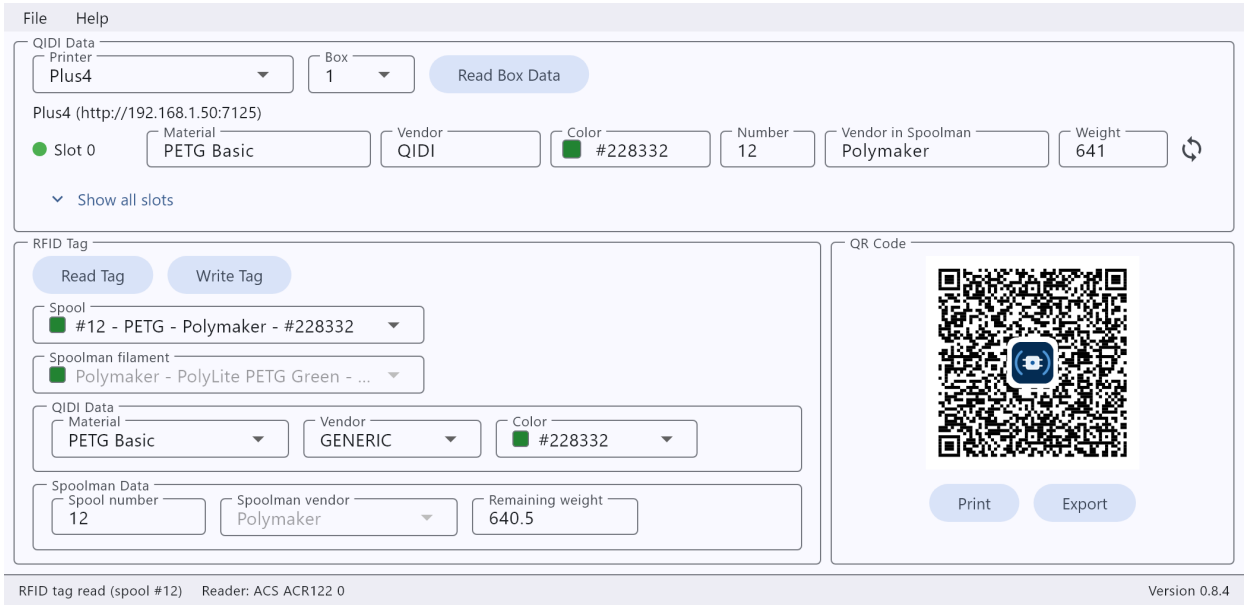


Figure 1: The complete main window with the three frames QIDI Data, RFID Tag and QR Code

1.2 Key terms

Term	Meaning
QIDI Box	The filament box on the printer with four slots for spools. A printer can have several boxes.
Slot	A position of a box, numbered 0 to 3 per box.
RFID tag	The small chip on the spool. RFID Wisp works with MIFARE Classic 1K.
Moonraker	The web service on the printer (Klipper). The app uses it to get the data of the QIDI box. Address, for example, `http://192.168.1.50:7125`.

Term	Meaning
Spoolman	A free server for managing filament spools (vendors, filaments, spools, remaining weight). Address, for example, `http://192.168.1.100:7912`.
Spool number	The number of the spool in Spoolman. It is stored on the tag (0 to 999). Without Spoolman it is simply a number you choose yourself.
Reader	An RFID reader connected to the PC (PC/SC, e.g. ACS ACR122). On Android the phone's NFC chip is the reader.
rfid_bridge	An optional add-on module for Klipper on the printer. It provides the spool number and the Spoolman vendor per slot and reports the active spool to Spoolman (chapter 11).

1.3 With and without Spoolman

You can use RFID Wisp **with Spoolman** or **without Spoolman**. This is decided in the settings ("Spoolman" tab) and noticeably changes the "RFID Tag" frame:

- **With Spoolman** the app looks up spools in Spoolman, creates new spools and writes the number Spoolman assigns to the tag. You can take material, vendor, color and weight from an existing Spoolman filament.
- **Without Spoolman** you choose material, vendor and color directly and assign the spool number yourself. The app then does not contact any Spoolman server at all.

All differences in detail are in section 6.6. If you do not use Spoolman, you can go straight to section 6.5 of chapter 6.

1.4 Conventions in this manual

- **Bold** text is a label from the app (buttons, fields, menus).
- Menu paths are written with an arrow: **File → Settings …**
- `Monospaced text` stands for addresses, file names and commands.
- Colored boxes highlight important points: **Note**, **Tip**, **Warning** and **Without Spoolman** (what is different in that mode).

2 Requirements and Installation

2.1 What you need

Platform	Requirements
Windows	64-bit Windows 11. Microsoft Visual C++ Redistributable (x64) – already present on most systems. A PC/SC-capable RFID reader (tested with the ACS ACR122) with its driver installed.
Linux	64-bit distribution with GTK 3 (`libgtk-3-0`), as recent as Ubuntu 24.04 or newer. The `pcscd` service and a PC/SC reader.
Android	Android 7.0 or newer. The phone's NFC chip must support MIFARE Classic (see chapter 9).

Optional, depending on what you want to use:

- the **Moonraker address** of your QIDI printer (for the "QIDI Data" frame and for the current material, vendor and color lists),
- a **Spoolman server** on your network,
- a **printer** set up on the computer for QR code labels (on Linux through CUPS). Saving as a PDF file works without a printer.

> **Note:** The app was tested with the QIDI Plus4 (firmware 1.8.2 and 2.1.1), the QIDI Max4 (firmware 01.01.06.05 and 01.01.07.01) and the QIDI Q2 (firmware 1.1.1) with one QIDI Box. Other firmware versions have not been checked.

2.2 Windows

You can find the current version at `https://github.com/ThorSc/RFIDwisp/releases/latest`. Two files are available for Windows:

- **`RFIDwisp-windows-x64-setup.exe`** (installer): double-click it and follow the steps. It creates a Start menu entry and, if you like, a desktop icon. A newer version can simply be installed over the existing one. To uninstall, use "Apps" in the Windows settings.
- **`RFIDwisp-windows-x64.zip`** (portable): extract it to a folder and run `rfid_wisp.exe`. Nothing is installed.

Windows warning on first start

The program file is not digitally signed yet. For that reason Windows 11 may object to it on the first start:

- If **"Windows protected your PC"** (SmartScreen) appears, click **More info → Run anyway**.
- If **Smart App Control** blocks the file completely (there is no "Run anyway" button), the only option for now is to turn Smart App Control off: **Windows Security → App & browser control → Smart App Control settings → Off**.

> **Warning:** Once turned off, Smart App Control may not be turned on again without reinstalling or resetting Windows. Think carefully before taking this step.

Even during the download the browser (Microsoft Edge here) may report that the file is not commonly downloaded. In the menu of the file choose “Keep”, then “Keep anyway” in the following dialog (steps 1 and 2 in the figure).

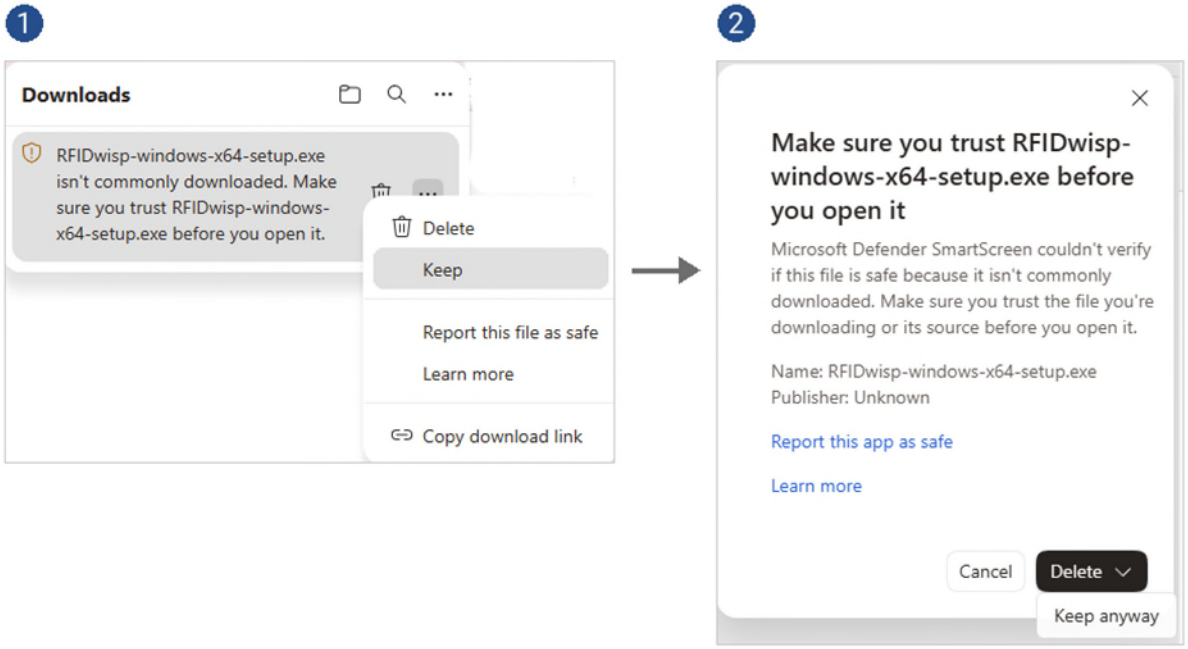


Figure 2: Browser warning at the download: “Keep”, then “Keep anyway”

2.3 Linux

- **Debian/Ubuntu**: `sudo apt install ./RFIDwisp-linux-x64.deb`. This adds a menu entry, and the command `rfid_wisp` is on the path. Remove it with `sudo apt remove rfid-wisp`.
- **Portable (any distribution)**: `tar xzf RFIDwisp-linux-x64.tar.gz`, then start `./RFIDwisp-linux-x64/rfid_wisp`.

For the reader to work, the smart card service must be running:

1. `sudo apt install pcscd` (this also installs `libpcsclite1`, which the app loads at runtime),
2. `sudo systemctl enable --now pcscd`,
3. if necessary, install the CCID/ACS driver for your reader in case the generic CCID driver does not recognize it.

With `pcsc_scan` (package `pcsc-tools`) you can check, independently of RFID Wisp, whether the reader is visible.

Two pitfalls with the ACR122U

If the reader shows up in `lsusb` but not in the app:

- **A kernel driver blocks the device.** The driver `pn533_usb` claims the reader, and `pcscd` then reports `Can't claim interface … LIBUSB_ERROR_BUSY`. Create `/etc/modprobe.d/blacklist-`

`pn533.conf` with the lines `blacklist pn533_usb`, `blacklist pn533` and `blacklist nfc`, run `sudo update-initramfs -u` and replug the reader (or reboot).

- **polkit denies access.** `pcscd` only serves users in an active local session. Over SSH or in a remote session `pcsc_scan` fails with "Access denied". Allow your user in `/etc/polkit-1/rules.d/99-pcscd.rules`: for the actions `org.debian.pcsc-lite.access_pcsc` and `org.debian.pcsc-lite.access_card` return `polkit.Result.YES`, then run `sudo systemctl restart polkit pcscd`.

2.4 Android

The app is not distributed through Google Play. For that reason Android asks for permission:

1. Download `RFIDwisp-android.apk` to the phone (or copy it there).
2. Open the file. If Android says that installing from this source is not allowed, open **Settings** in the message and allow it for the app you used to open the file (browser or file manager).
3. Confirm the installation. Play Protect may ask for a check because the app does not come from the Play Store.

To update, simply install a newer APK over the old one; the settings are kept. This works from version 0.3.1 on. The APKs up to 0.3.0 were each signed with a different key, which Android does not accept as an update: if you have such a version, uninstall it once before installing 0.3.1 or newer (the settings are lost this one time).

2.5 Setting up the reader (Windows and Linux)

1. Connect the reader to a USB port. On **Windows** the manufacturer's driver must be installed; smart card support (WinSCard) is part of Windows. On **Linux** `pcscd` must be running (section 2.3).
2. Start RFID Wisp and open **File → Settings … → Reader**.
3. Select your device in the list and tick **Use as default reader** (section 3.4).
4. Confirm with **OK**. The status bar now shows "Reader: <name>".

3 First Start and Settings

3.1 Starting the app

On start the app briefly shows a **splash screen** with the logo. In the background it checks whether the default reader is connected and usable and (unless switched off) looks for a new version. Then the main window appears.

On the very first start no printers are set up, no reader is chosen and no Spoolman address is saved. The main window tells you so: the “QIDI Data” frame says “No printer configured. Add one under File → Settings.”, and the “RFID Tag” frame says that no default reader is chosen. So set up the settings first.

3.2 The settings dialog

Open **File → Settings …**. The settings are spread over tabs:

Tab	Contents	Available on
General	Language, update check, export and import of the settings	all
Reader	Choice of the RFID reader	Windows and Linux only
Printers	Your QIDI printers (name and Moonraker address)	all
Spoolman	Address of the Spoolman server and the “Use Spoolman” switch	all

> **Note:** Nothing is saved until you click **OK**. **Cancel** discards all changes on all tabs. If a printer name or an address is invalid, **OK** jumps to the faulty field and marks it (“Required” or “Invalid address, e.g. http://192.168.1.50:7125”).

3.3 “General” tab

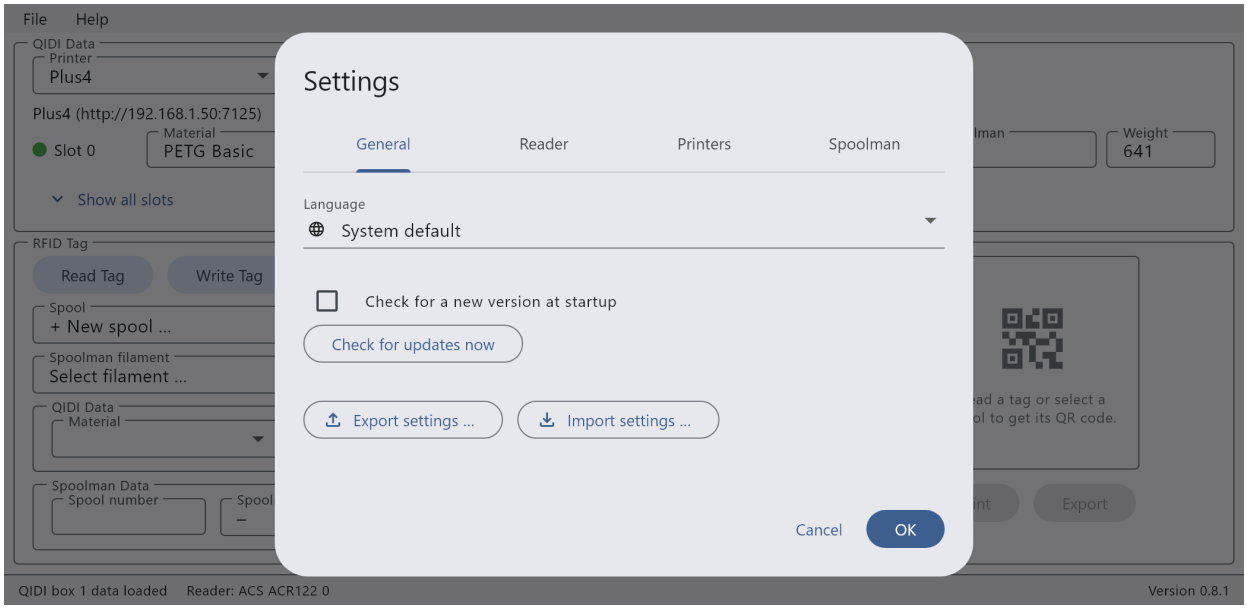


Figure 3: Settings dialog, “General” tab

- **Language**: System default, German, English or Ukrainian. With “System default” the app follows the language of Windows, Linux or Android. In the list each language is shown with its flag.
- **Check for a new version at startup**: switches the automatic update check on or off (chapter 10).
- **Check for updates now**: checks immediately. You will see “You are using the latest version.”, an error message (“The update check failed.”) or the offer to install the new version.
- **Export settings …** and **Import settings …**: see section 3.7.

3.4 “Reader” tab (Windows and Linux only)

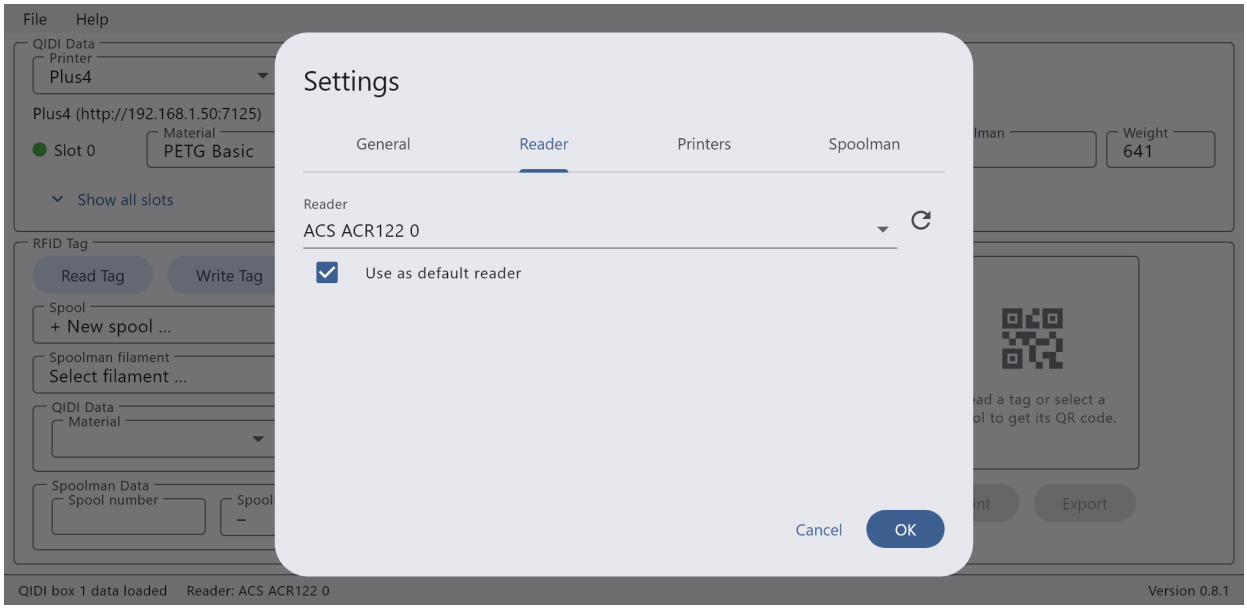


Figure 4: “Reader” tab with a device that was found

Here you see the RFID readers the computer has found. The app **tests the capabilities of each device itself** instead of relying on the name. Other smart card devices (for example a YubiKey) are **never opened** and do not appear in the list.

- Choose the device you want in the **Reader** list.
- The arrow icon (**Search again**) refreshes the list, for example after you have plugged in a device. While it searches you see “Searching for readers …”. If the app finds nothing, it says “No RFID reader found”.
- Tick **Use as default reader** so the app uses this device. There can only be one default reader.
- A saved default reader that is currently not connected appears in the list as “<name> (not connected)”.

What happens without a default reader?

Without a default reader the **Read Tag** and **Write Tag** buttons are disabled, and the “RFID Tag” frame says: “No default reader chosen. Choose one under File → Settings.” All other parts of the app (QIDI Data, QR code export) keep working.

The check at startup

At startup the app checks whether the default reader can be reached. The result is shown in the status bar next to the status message:

Status bar	Meaning
Reader: <name>	Everything is fine. A reader with no tag on it counts as usable, because RFID capability can only be verified with a tag present.
Checking reader <name> …	The check is still running.
Default reader not found: <name>	The device is not connected or not reachable (red).
Default reader cannot read RFID tags: <name>	The device is not an RFID reader (red).
Reader <name>: the tag on it is not a readable MIFARE Classic 1K	A tag that QIDI printers cannot use is on the device (red).
Smart card service not available	Windows: the “Smart Card” service is not running. Linux: `pcscd` is not installed or not started (red).

3.5 “Printers” tab

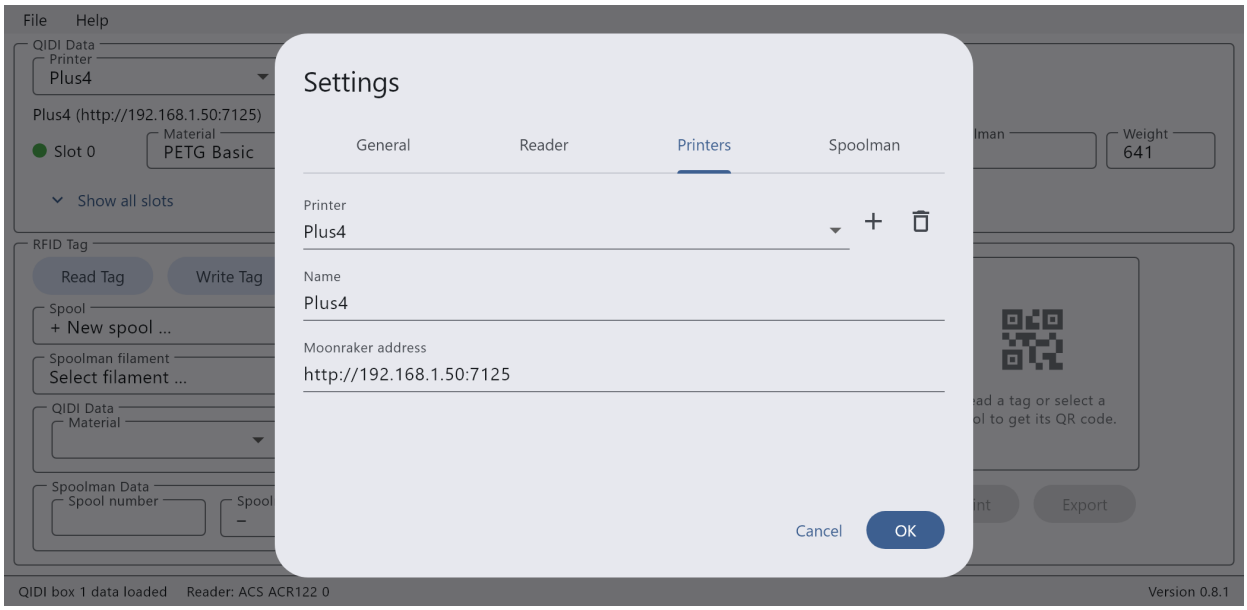


Figure 5: “Printers” tab with one printer set up

Here you set up the QIDI printers whose boxes you want to read. There is no upper limit on the number.

1. Click **+** (**New printer**).
2. Enter a **Name**, for example “Plus4 Workshop”. The name is required.
3. Enter the **Moonraker address**, for example `http://192.168.1.50:7125`. The app adds a missing `http://` itself. This address is required as well.
4. To delete a printer, select it in the list and click the trash can icon (**Delete printer**).

To change an existing printer, select it in the list and edit its name or address. With no printer set up the list says “No printers yet”.

> **Tip:** You can also find the Moonraker address in the address bar of the printer's web interface (Fluidd or Mainsail). Moonraker's default port is 7125.

The printers serve two purposes: they provide the “QIDI Data” frame with the box data, and they provide the **lists of materials, vendors and colors** offered by the “RFID Tag” frame (section 6.2).

3.6 “Spoolman” tab

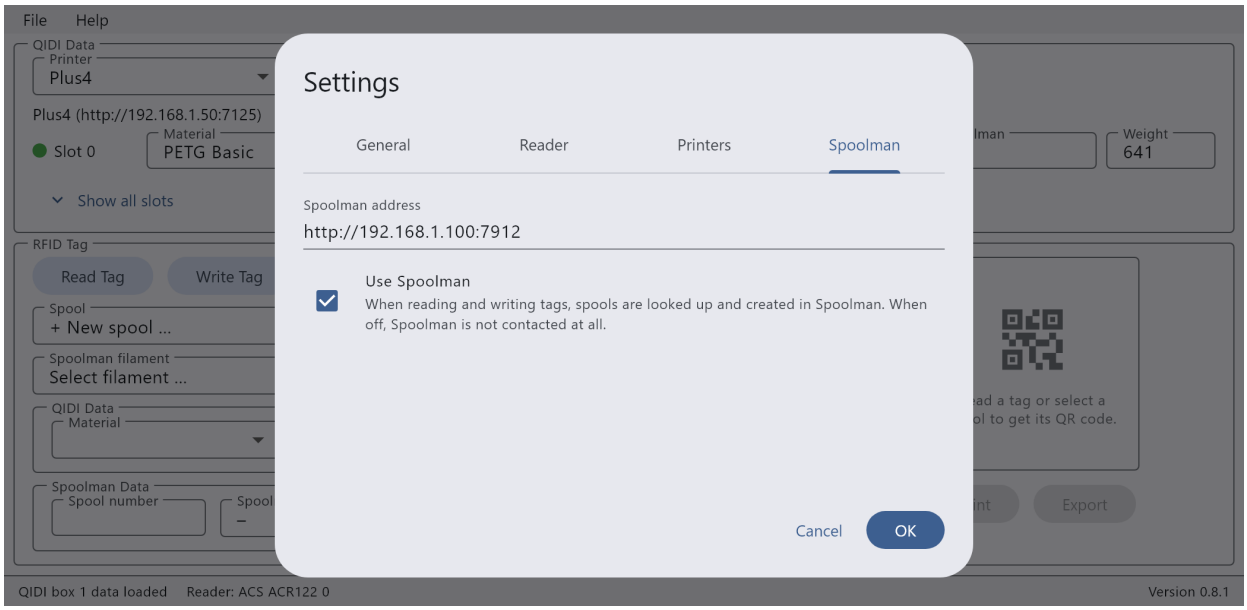


Figure 6: “Spoolman” tab with address and switch

- **Spoolman address**: for example `http://192.168.1.100:7912`. A missing `http://` is added.
- **Use Spoolman**: When reading and writing tags, spools are looked up and created in Spoolman. When off, Spoolman is not contacted at all.

> **Without Spoolman:** The app uses Spoolman in the “RFID Tag” frame only if **both** are true: an address is entered **and** “Use Spoolman” is ticked. If you leave the address empty, the dialog shows the option as switched off the next time you open it. How the app behaves then is described in sections 6.5 and 6.6. This also applies to the “QIDI Data” frame: with “Use Spoolman” switched off, the app does not ask Spoolman there for vendor names and remaining weight, even if an address is still saved.

If the Spoolman setting changes (address or switch), the app reloads the Spoolman data and **clears the form in the “RFID Tag” frame**. Entries that have not been written are lost.

3.7 Exporting and importing settings

On the “General” tab you can save all settings (language, update check, reader, printers, Spoolman) to a file and restore them. This is handy as a backup or to have the same settings on another computer or phone.

- **Export settings …** saves everything to a JSON file whose location you choose in the save dialog. The app then reports “Settings saved to <file>.”
- **Import settings …** loads the values from such a file. The import **only fills in the dialog**; like any other change it takes effect only with **OK** and is discarded by **Cancel**. Settings that are missing from the file stay unchanged.

Message during import	Meaning
The file is not a valid JSON file.	The file is damaged or not a JSON file.

Message during import	Meaning
The file is not an RFID Wisp settings file.	Valid JSON, but not exported by this app.
The settings file was written by a newer version of RFID Wisp.	Update the app and try again.
The file could not be read or written: …	A permission or disk problem.

4 The Main Window

4.1 Layout

From top to bottom the main window consists of:

- the **menu bar** (**File** and **Help**),
- the **QIDI Data** frame (chapter 5),
- below it the **RFID Tag** frame (chapter 6) and – on Windows and Linux – to its right the **QR Code** frame (chapter 7). The QR Code frame is one third of the combined width and as high as the RFID Tag frame,
- the **status bar** (section 4.3).

The window height adapts to the content when it changes (for example when all slots are shown). If space gets tight, the content can be scrolled. The “RFID Tag” and “QR Code” frames exist only where a reader is supported; the phone has no QR Code frame.

4.2 The menu

Menu	Entry	Effect
File	Settings …	Opens the settings dialog (chapter 3).
File	Exit	Closes the app. Only available on Windows and Linux.
Help	Open log	Opens the log file in the Windows editor. Currently only available on Windows (section 12.3).
Help	About RFID Wisp	Shows version, copyright and license information. The licenses of the packages used can be viewed with the licenses button.

4.3 The status bar



Figure 7: Status bar at the bottom of the window with message, reader and version

From left to right the status bar at the bottom shows:

- the **latest message** of the app, for example “QIDI box 1 data loaded” or “RFID tag written (spool #12)”. At startup it says “Ready”,
- the **state of the reader** (section 3.4), in red if there is a problem,
- if there is a newer version, the underlined link **“New version … available – download”** (chapter 10),
- at the far right the **version number** of the app.

If a message is too long for the bar, it is shortened with “…”. If you hold the mouse pointer over it, a tooltip shows the full text.

4.4 Busy pointer and tooltips

- While the app is checking a printer, reading a box, reading or writing a tag, or printing or saving a QR code, the mouse pointer turns into the busy ring over the **whole window**. Clicks still go through; the affected fields and buttons are disabled in the meantime.
- Almost every field has a **tooltip**: hold the mouse pointer over it for a moment and a hint explains what the field is for.

4.5 What works when – overview

The app is built so that each part also works without the others:

If missing …	… then
Printer (none set up)	"QIDI Data" only shows the hint "No printer configured". Reading and writing tags keeps working. The material, vendor and color choices then come from the **list built into the app** instead of from a printer.
Printer not reachable	"QIDI Data" reports in red "<name> (<address>) – not connected" with an arrow to check again. Reading and writing tags keeps working, with the built-in list.
Reader (no default reader)	"Read Tag" and "Write Tag" are disabled. "QIDI Data" and the QR code of a spool that is already shown keep working.
Spoolman (switched off or no address)	The "RFID Tag" frame works with values you choose and a spool number you assign yourself (section 6.5).
Spoolman not reachable	The Spoolman lists stay empty, and the status bar says so. See section 6.7.
Moonraker / `[spoolman]` component / rfid_bridge	In "QIDI Data" the spool number and the Spoolman vendor stay empty (chapters 5 and 11).
Paper printer	"Print" in the QR Code frame is disabled. "Export" as PDF still works.

5 The “QIDI Data” Frame

This frame shows what a QIDI box has reported for each of its slots. It only needs the Moonraker address of the printer. It **does not write anything** on tags; apart from re-reading a slot (section 5.2) it is a pure display.

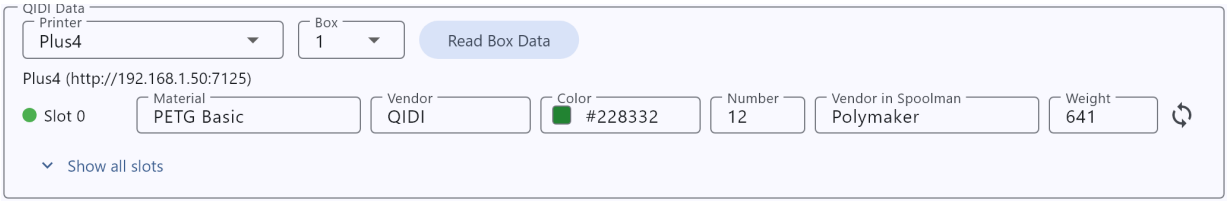


Figure 8: “QIDI Data” frame with one loaded slot (default view)

5.1 Choosing printer and box

1. Choose a **Printer** (from the ones in the settings). If there is only one, it is already selected. The app remembers the printer you chose last.
2. At startup and whenever you change the printer, the app asks the printer **how many QIDI boxes** it has and reads the first one. Meanwhile it says “Checking printer …” with a small progress ring.
3. Choose the box you want under **Box** (1, 2, …). The list is empty as long as the printer cannot be reached.
4. Click **Read Box Data** to read the selected box. A box is read automatically when you switch to it.

Below the selection fields you see the state of the printer, for example “Plus4 (http://192.168.1.50:7125)”. If the app could not reach the printer, it says instead, in red, “… – not connected”, and the status bar names the reason (“QIDI box data not available: …”). The **arrow** next to it (**Check the printer again**) tries once more, for example after you have switched on the printer. If the printer reports its box as switched off (`enable_box = 0`, for example after a print from the external spool), the app says so below it in red, and the box list stays greyed out. Switch the box on again on the printer, then check again with the arrow.

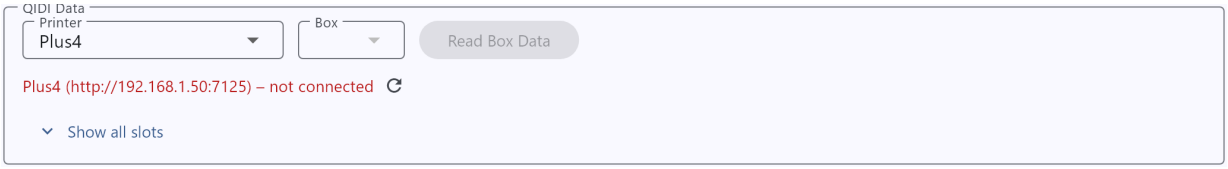


Figure 9: Red notice “… – not connected” with an arrow to check again

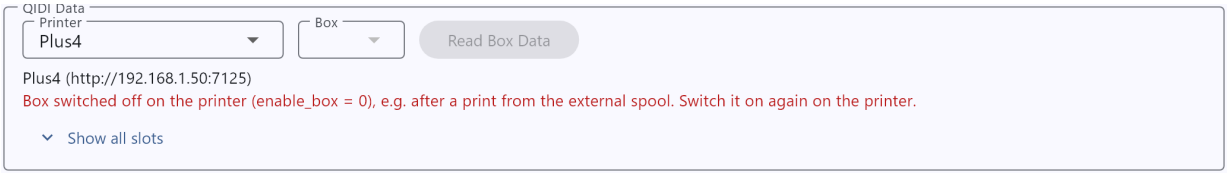


Figure 10: Notice “Box switched off on the printer”

If reading a box fails, the status bar says “Could not read the QIDI box: …”. If you pressed **Read Box Data** yourself, the “QIDI box error” dialog with the error message opens as well.

5.2 The slots

By default only **slot 0** of the box is visible. **Show all slots** shows all four slots; **Hide slots 1-3** switches back. Within each box the slots are named 0 to 3. For box 2 these are internally slots 4 to 7, but the app still labels them 0 to 3.

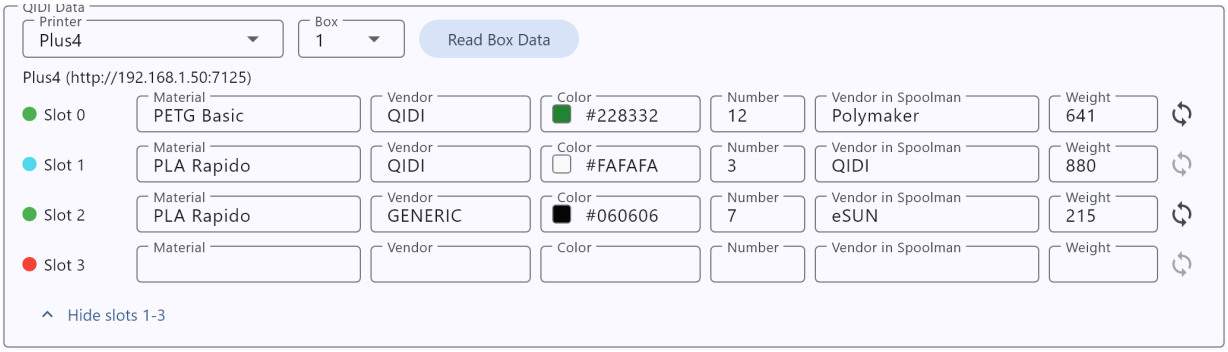


Figure 11: "QIDI Data" frame with all four slots

Field	Meaning	Source
Dot in front of the slot	**Red**: slot empty. **Green**: filament loaded. **Cyan**: loaded and currently feeding the extruder.	Printer
Material	Filament type the box has recognized.	Printer (from the RFID chip)
Vendor	Filament vendor according to QIDI.	Printer (from the RFID chip)
Color	Filament color as `#RRGGBB` with a color swatch.	Printer (from the RFID chip)
Number	Spoolman spool number stored on the tag.	`rfid_bridge` on the printer
Vendor in Spoolman	Name of the Spoolman vendor that belongs to the spool.	`rfid_bridge` plus Spoolman address
Weight	Remaining weight of the spool in whole grams.	Spoolman (address required)

The app takes the names for material, vendor and color from the printer's own filament list. Only where this list cannot be read does it use the built-in list.

Why are "Number", "Vendor in Spoolman" or "Weight" empty?

- **Number** and **Vendor in Spoolman** come from the optional Klipper module `rfid_bridge` (chapter 11). Without this module both fields stay empty. On the **QIDI Q2** the module is the only source of spool numbers.
- **Vendor in Spoolman** and **Weight** also need a **Spoolman address** in the settings and the "Use Spoolman" option switched on. Otherwise they stay empty.
- **Weight** is also empty if the slot has no spool number or Spoolman does not know the spool.
- The weight is fetched from Spoolman again when you click **Read Box Data**, switch to another box or select a printer. The value is rounded to whole grams; the exact weight is shown in the "RFID Tag" frame and in Spoolman.

> **Without Spoolman:** Without a Spoolman address, or with "Use Spoolman" switched off, "QIDI Data" only shows material, vendor, color and (with `rfid_bridge`) the number. The "Vendor in Spoolman" and "Weight" columns stay empty. That is not an error.

Re-reading a slot

To the right of every loaded slot that is not currently feeding the extruder there is a symbol with two arrows. It makes the box read the tag of this slot once more. Use it when a slot shows the data of its neighbour after a start or has no number (section 11.2).

After a confirmation the app sends the command `RFID_READ SLOT=<n>` and, if the printer knows it, then `READ_RFID SLOT=<n>` to the printer through Moonraker. Depending on the firmware only one of the two commands reads (QIDI Plus4 with firmware 2.1.1: `RFID_READ`, QIDI Max4 with firmware 01.01.07.01: `READ_RFID`); the other has no effect. The box moves the filament of this slot back and forth, which takes about half a minute. Then the app reads the box again.

> **Note:** This only works while the printer is idle: it must not be printing, and no filament may be loaded into the extruder. During a print the app refuses to re-read. `rfid_bridge` then reports the slot that was read again as the active spool.

5.3 Warning “The printer has no connection to Spoolman”

If Moonraker on the printer reports that it cannot reach Spoolman (for example after a network interruption), a **red warning** with a warning triangle appears in the frame; the same message appears in the status bar. The consequence: **spool changes of the box are then not reported to Spoolman**, so the active spool in Spoolman does not follow what happens in the box. The app checks this when the printer is checked and when you press **Read Box Data**. Make sure the printer can reach Spoolman again (network, Spoolman server, the `[spoolman]` section in `moonraker.conf`) and press **Read Box Data** to refresh the warning.

6 The “RFID Tag” Frame

In this frame you read and write the RFID tag that is on the default reader (on Android: the phone's NFC chip, chapter 9). It is the most important part of the app.

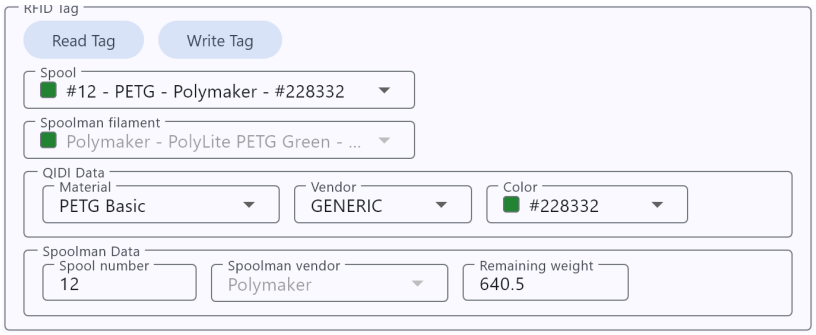


Figure 12: “RFID Tag” frame with Spoolman switched on

6.1 Layout of the frame

From top to bottom the frame contains:

- the **Read Tag** and **Write Tag** buttons,
- **with Spoolman only:** the **Spool** and **Spoolman filament** selections,
- the **QIDI Data** group with **Material**, **Vendor** and **Color** – this is what the QIDI box reads from the tag. **Without Spoolman** this group also contains the **Spool number** field,
- **with Spoolman only:** the **Spoolman Data** group with **Spool number**, **Spoolman vendor** and **Weight**.

The data always goes into a fixed sector of the tag (sector 1), because that is where the QIDI box expects it. This sector cannot be changed.

6.2 Where do the selection lists come from?

The lists for **Material**, **Vendor** and **Color** come from the filament tables of your **configured printers**. The app **merges the tables of all printers** so that a written tag is understood in **every** one of your boxes. If no printer could be read, it uses the list built into the app (matching the QIDI firmware) until a printer is reachable.

- The colors are a **fixed palette** from QIDI. So you choose the QIDI color closest to the color of the spool, not just any color.
- The vendor “GENERIC” (“Generic” on some printers) stands for “no particular vendor” and is the default.
- The lists load at startup and when the printer list or the selected printer changes. If you have already chosen something, a list that finishes loading later no longer changes your selection.

6.3 Reading a tag

1. Put the tag on the reader. On **Android** first tap **Read Tag** and then hold the tag to the back of the phone (chapter 9).
2. Click **Read Tag**. The fields show the contents of the tag; the status bar reports “RFID tag read (spool #12)”.

What happens depends on the contents of the tag:

Tag	Result
Written (by RFID Wisp or the QIDI box)	Material, vendor, color, spool number and the weight stored on the tag are shown.
Blank (zeros only)	The form is blank; the status bar says “Blank RFID tag detected. Ready to write new spool data.”
Written by something else (values that appear in no table)	An error dialog “Read RFID tag” says that the values cannot be matched and that the tag was probably not written by this application. The form stays unchanged.
Not a MIFARE Classic 1K	An error message says that the tag is not the type QIDI printers require.
Locked (other key)	Error message “Cannot authenticate to sector …: the tag uses a key this app does not know.”
Not detected	“No RFID tag detected on the reader.”

Taking the tag off clears the fields

On Windows and Linux, after reading the app asks the reader **once a second** whether the tag is still on it. If you take it off, the fields are cleared and the status bar reports “RFID tag removed, fields cleared”. This way you never mistake the display of a previous tag for the new one. On **Android** this does not exist (the phone cannot detect it); the fields stay.

> **Note:** After **writing**, the fields are **not** cleared, even if you take the tag off. This is intentional (section 6.8).

What happens with Spoolman when reading?

If Spoolman is active, the app shows the tag immediately and then asks Spoolman again **in the background**, because the spool may have been created there after the app was started. If Spoolman knows the spool number of the tag, the spool is selected in the **Spool** field; spool number, Spoolman vendor and the **remaining weight** come from Spoolman. **Material, vendor and color stay exactly the values from the tag** (the app does not derive them from Spoolman again, otherwise “PETG Rapido” would silently become “PETG”).

While Spoolman is still answering, **Write Tag** is disabled for a moment. This prevents a new spool from being created by accident although Spoolman actually knows the spool of the tag.

If Spoolman does **not** know the number, the entry **+ New spool …** stays in the **Spool** field and the number from the tag stays in the **Spool number** field. What that means when writing is explained in section 6.7.

6.4 Writing a tag with Spoolman

Prerequisite: Spoolman is set up and switched on (section 3.6). There are two ways: create a **new spool**, or write an **existing spool** to a tag again.

A) Creating and writing a new spool

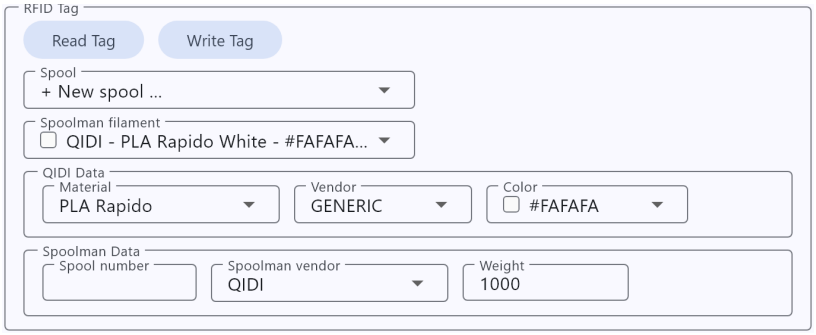


Figure 13: Filled-in form for a new spool before writing

1. Leave the entry **+ New spool …** in **Spool**.
2. Choose a **Spoolman filament** from Spoolman. The list is sorted by vendor, material and name and shows color swatch, vendor, name, color and weight for each entry. The app prefills **Material**, **Spoolman vendor**, **Color** and **Weight** from it.
   Alternatively choose only a **Spoolman vendor** and let the app find the filament that matches material, vendor and color.
3. **Check the prefilled values.** Spoolman material names are free-form; the app picks the closest QIDI material and the closest QIDI color. Adjust **Material**, **Vendor** (QIDI) and **Color** if needed. You can also change the **Weight** (initial weight in grams, 0 to 10000).
4. Put the tag on the reader (on Android: hold it to the phone after clicking the button) and click **Write Tag**.

The following happens, in order:

1. The app **creates the spool in Spoolman** (with the initial weight, if one is entered). Spoolman assigns the number.
2. The form immediately shows the new spool as the selected **Spool**, and the spool number is filled in.
3. The app **writes** material, vendor, color, weight, the Spoolman vendor and the spool number **to the tag** (sector 1).
4. It **reads the tag back to verify**. Only when that matches does the status bar report “RFID tag written (spool #12)”.

> **Tip:** Because the form switches to the **new spool** after it has been created, writing again (for example because the tag was not placed correctly) does **not** create a second spool. Further copies of the same tag do not create new spools either.

If **neither a filament nor a Spoolman vendor** is selected for a new entry, the app aborts: “Select a Spoolman vendor or filament to create the spool in Spoolman.” If Spoolman has no filament for the combination of material, vendor and color, it reports: “No matching filament found in Spoolman for this material/vendor/color combination.” Then create the filament in Spoolman or directly choose a **Spoolman filament**.

B) Writing an existing spool to a tag

You need this when a tag is defective, when you want a second tag for the same spool or when you want to adjust a tag afterwards.

1. Open the **Spool** list and choose the spool. The entries look like this: “#12 - PETG - Polymaker - #FF0000” (number, material, vendor, color) with a color swatch, and the **newest spool is at the top**.
2. The fields fill in: material and color are derived from the Spoolman filament (closest QIDI material and QIDI color), the Spoolman vendor and the number come from Spoolman. **Check material, vendor and color** and change them if the derivation is not right.
3. The **Remaining weight** field is **for display only** and shows what Spoolman currently has stored for the spool. The **Spoolman filament**, the **Spoolman vendor** and the weight are disabled; they belong to the existing spool and cannot be changed here.
4. Put the tag on the reader and click **Write Tag**. **Nothing is created or changed in Spoolman**; only the tag is written. The remaining weight is written to the tag as the “last weight”.

> **Note:** To create a new spool again, choose **+ New spool …** in **Spool** once more. The form is cleared in the process.

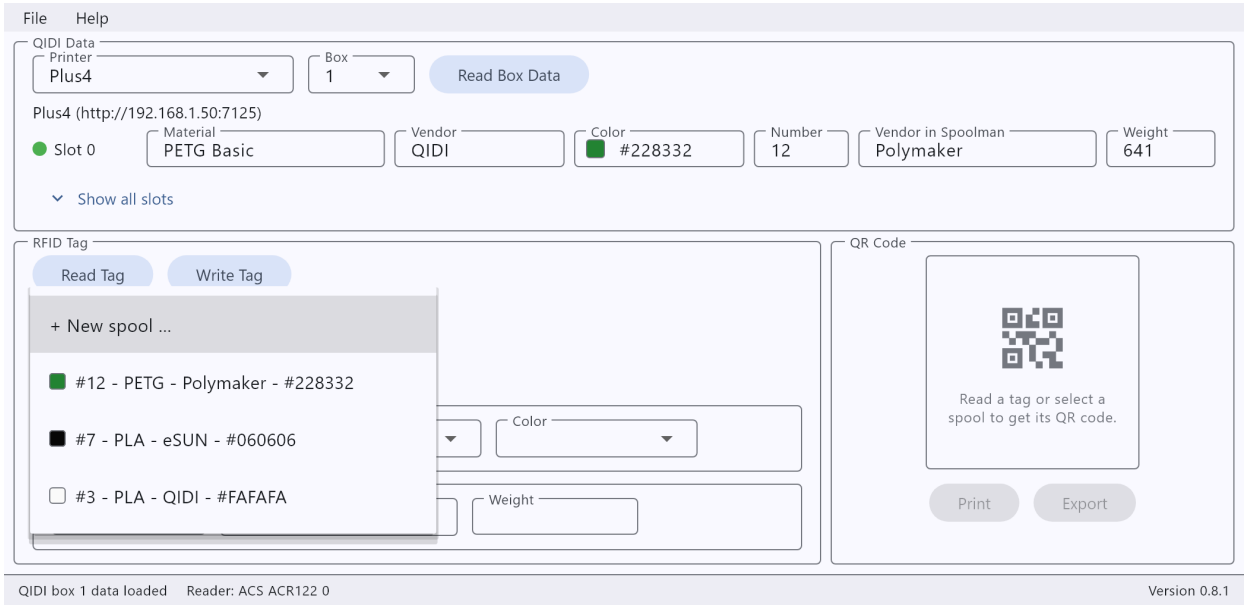


Figure 14: Open “Spool” list with color swatches

When do I see new spools from Spoolman?

The app loads vendors, filaments and spools from Spoolman at startup, when you change the Spoolman settings and after every **Read Tag**. If you create a new spool in Spoolman in the browser while the app is running, it appears in the list after the next **Read Tag** (or after restarting the app).

6.5 Writing a tag without Spoolman

> **Without Spoolman:** This section applies when **Use Spoolman** is switched off or no Spoolman address is entered.

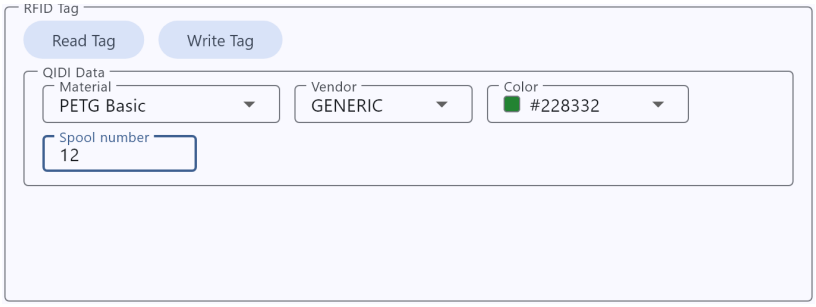


Figure 15: "RFID Tag" frame without Spoolman: only the "QIDI Data" group with spool number

Without Spoolman the frame is deliberately slim. There is **no** "Spool" selection, **no** "Spoolman filament" and **no** "Spoolman Data" group. What remains is the **QIDI Data** group with **Material**, **Vendor**, **Color** and – added here – the **Spool number** field.

1. Choose **Material**, **Vendor** and **Color**.
2. Enter the **Spool number**: 0 to 999, digits only, at most three digits. This is the number by which your QIDI box (and, if used, `rfid_bridge`) recognizes the spool. You assign it yourself.
3. Put the tag on the reader and click **Write Tag**.

What is different:

- **There is no contact with Spoolman.** No spool is created, and the number is not checked against Spoolman.
- **You are responsible for unique numbers yourself.** The app does not check whether you have already used the number for another spool.
- If the spool number is missing, "Please enter a spool number." appears. If material or color is missing, "Please select material, vendor and color before writing." appears.
- There is **no weight field**. The weight 0 is written to the tag, and the Spoolman vendor entry on the tag stays empty (0).
- When **reading** a tag, the app shows material, vendor, color and spool number. The weight and the Spoolman vendor of the tag are not shown (the fields do not exist).
- The vendor field refers exclusively to the **QIDI vendor** (such as "GENERIC"); there is no Spoolman vendor.

> **Tip:** If you do not use Spoolman, you probably just want the QIDI box to recognize filament from other vendors correctly. Material, color and any spool number are enough for that. For example, use a consecutive number for each spool and note it down (or print the QR code for it).

6.6 With and without Spoolman compared

	With Spoolman	Without Spoolman
Prerequisite	Address entered **and** "Use Spoolman" on	Option off **or** address empty
"Spool" selection	yes (existing spools, or "+ New spool …")	no
"Spoolman filament" selection	yes (new spool only)	no

	With Spoolman	Without Spoolman
“Spoolman Data” group	yes: spool number, Spoolman vendor, weight	no
Spool number	comes from Spoolman, display only	you enter it yourself (0–999)
Create a new spool in Spoolman	yes, automatically when writing	no
Weight	initial weight (new spool) or remaining weight (existing spool), goes on the tag	no field; the tag gets 0
Spoolman vendor on the tag	yes	no (empty)
Network access to Spoolman	yes	no
Checking the spool number	Spoolman assigns and knows the number	none, you are responsible
QR code appears …	when material, color and number are known; for a new spool therefore only **after writing**	as soon as material, color and number are entered
QR code and PDF	with Spoolman vendor, color and material	Spoolman lines empty

6.7 Special cases and errors when writing

Spoolman is switched on but not reachable

If the app cannot reach Spoolman (wrong address, server off, network problem), the lists stay empty and the status bar reports “Could not load data from Spoolman: …”. You can still choose material, vendor and color. When writing a **new spool** the following happens:

- If there is **no number** in the form, the app aborts and shows Spoolman's error message in a dialog.
- If there is a number in the form (for example because you read a tag before), **the app writes the tag with this number** and reports “Spoolman sync skipped: …” in the status bar. No new spool is created then.

> **Tip:** If you do not need Spoolman right now, switch it off on the “Spoolman” tab and write the tag as described in section 6.5.

The tag that was read belongs to a spool Spoolman does not know

This happens when the spool was deleted in Spoolman or the tag comes from another Spoolman installation. After reading, **+ New spool …** is shown in the “Spool” field and the tag's number is in the “Spool number” field. If you write now:

- If a **Spoolman vendor** is selected (it is prefilled from the tag if Spoolman knows it) or a **Spoolman filament**, the app creates a **new spool** and writes its **new number** to the tag.
- If neither is the case, it writes the tag with the **previous number** (the app cannot create the spool anew).

Other messages when writing

Message	Cause and remedy
Please select material, vendor and color before writing.	At least one of the fields is still empty. Choose material and color.
Please enter a spool number.	Without Spoolman only: the “Spool number” field is empty.
No spool selected.	With Spoolman but no number in the form although no new spool is to be created. Choose a spool.
The weight must be a number.	The “Weight” field contains something other than a number.
The weight must be between 0 and 10000 grams.	Correct the weight.
Spool number must be an integer between 0 and 999	Spoolman has assigned a spool number above 999. The tag can only store 0 to 999 (Appendix A). Create the spool in Spoolman with a smaller number or work without Spoolman.
Writing the tag failed.	The tag was not held still or close enough. Place it again and retry, and if necessary test another tag.

6.8 Writing several tags in a row

After writing, **all values stay in the form**, even if you take the tag off. This way you can write the **next tag** with the same data right away, for example when you need several labels or spare tags for one spool:

1. Write the tag (status bar: “RFID tag written (spool #12)”).
2. Take the tag off and put the next one on.
3. Click **Write Tag** again. With Spoolman no further spool is created, because the form already shows the spool that was created.

If you want to continue with something completely new, choose **+ New spool …** in **Spool** (with Spoolman); that clears the form.

> **Warning:** The form values are replaced by the contents of a tag when you **read** it. If you want to write the same tag several times, do not read another tag in between.

7 The “QR Code” Frame

The QR Code frame is to the right of the RFID Tag frame (Windows and Linux only, not on Android). It shows the QR code of the spool that the “RFID Tag” frame is currently showing and can turn it into a label.



Figure 16: “QR Code” frame with code and the “Export” and “Print” buttons

7.1 When does the code appear?

The code appears as soon as the “RFID Tag” frame shows a **complete spool**: **material**, **color** and **spool number**. This is the case:

- after **reading** a tag,
- after **choosing a spool** in Spoolman,
- after **writing** a new spool (only then does it have a number),
- **without Spoolman**, as soon as you have entered material, color and a spool number.

Before that, the frame says “Read a tag or select a spool to get its QR code.” The code changes live when you change values in the form. The logo of the app sits in the middle of the code; the highest error correction level makes sure it can still be read.

7.2 What is in the code?

The code contains text, one entry per line (the names are the same in every language):

```
Number: 12
QIDI Material: PLA
QIDI Vendor: GENERIC
QIDI Color: #FF0000
Spoolman Vendor: Polymaker
Spoolman Color: #FF0000
```

Without Spoolman: The two Spoolman lines stay empty (“Spoolman Vendor:” and “Spoolman Color:”), and the Spoolman details next to the code are missing in the PDF as well.

7.3 Exporting and printing

Both produce the same **A4 page** (DIN A4): the QR code is 20 × 20 mm and sits at the top left. To its right, one below the other in 14 pt and surrounded by a light frame, are: the **spool number**, the **Spoolman material** and the **Spoolman vendor**. A line that is too long for the page is shortened with "…".



Figure 17: Example of the exported A4 page with QR code and spool details

- **Export**: creates the PDF file and opens your system's save dialog. There you choose folder and name; "qr-code-spool-<number>.pdf" is suggested. If you cancel the dialog, nothing happens. After saving, the status bar reports "QR code of spool #12 saved: <path>".
- **Print**: prints the page on the system's **default printer** (on Linux through CUPS). The status bar reports "QR code of spool #12 sent to printer <name>".

Print is disabled as long as no printer is available (tooltip: "No printer available."). The app looks again when you activate its window, for example after you have connected a printer. **Export** always works.

> **Tip:** Print the page on label paper or cut out the area with the code and stick it on the spool. A phone camera or a scanner can read the code and shows the text above.

If printing or saving fails, a dialog "Print QR code" or "Export QR code" appears with the cause ("No printer available.", "Printing failed." or "The PDF file could not be saved: …").

8 Typical Workflows Step by Step

8.1 First-time setup

1. Install and start the app (chapter 2).
2. Open **File → Settings …**.
3. **Reader** tab: choose the device and mark it as the default (Windows/Linux).
4. **Printers** tab: create a printer with **+**, enter name and Moonraker address.
5. **Spoolman** tab: enter the address and tick **Use Spoolman** – or leave the address empty if you do not use Spoolman.
6. **OK**. The status bar must say “Reader: <name>”, and “QIDI Data” shows the first box of the printer.

8.2 Registering new filament from another vendor with Spoolman

1. Create the filament in Spoolman (if it does not exist yet).
2. In RFID Wisp leave **+ New spool …** in **Spool** and choose the filament under **Spoolman filament**.
3. Check and adjust **Material**, **Vendor** and **Color** (QIDI), and check the **Weight**.
4. Put a blank tag on the reader, **Write Tag**.
5. Optional: in the “QR Code” frame **Print** or **Export** and stick the label on the spool.
6. Put the spool with the tag into the QIDI box. With **Read Box Data** you check whether the box has recognized material and color correctly.

8.3 Registering new filament without Spoolman

1. Switch Spoolman off on the “Spoolman” tab or leave the address empty.
2. Choose **Material**, **Vendor** and **Color** and enter a free **Spool number**.
3. Put a blank tag on the reader, **Write Tag**.
4. Insert the spool and check with **Read Box Data**.

8.4 Finding out what is on a tag

1. Put the tag on the reader.
2. **Read Tag**.
3. Look at the fields and, if present, the QR code. Taking the tag off clears the fields again.

8.5 Replacing the tag of an existing spool

1. Choose the spool in **Spool**.
2. Check material, vendor and color.
3. Put the new tag on the reader, **Write Tag**. Nothing changes in Spoolman.

8.6 Creating a QR label for a spool in the box

1. Take the spool out of the box and put it on the reader (or read the tag).
2. **Read Tag**: the QR code appears.
3. **Export** or **Print**.

8.7 Checking which spool is where

1. In the “QIDI Data” frame: choose printer and box, **Show all slots**, **Read Box Data**.
2. The **dot** shows which slots are occupied and which one is currently feeding the extruder; **Weight** shows (with Spoolman) the remaining weight.

9 Using the App on Android

On Android the app uses the **NFC chip of the phone** as the reader. The operation is the same as on the PC, with these differences:

Difference	Explanation
No “Reader” tab	There is nothing to choose; the phone is the reader.
No QR Code frame	There is no room on the phone, and PDFs cannot be saved there. You print and export the labels on the PC.
No “Exit” menu entry	Android ends apps by itself.
Put the tag on only after tapping	First tap **Read Tag** or **Write Tag**; the status bar then says “Hold the RFID tag to the back of the device …”. Hold the tag to the **back** of the phone until the action is finished.
Fields stay after reading	The phone cannot detect when the tag is taken away.



Figure 18: Main window on the phone

9.1 Setting up NFC

- **NFC must be switched on** (Android settings). Otherwise the app says “NFC is switched off. Turn it on in the device settings.” and in the status bar “Reader … is switched off. Turn on NFC in the device settings.”
- The NFC chip must support **MIFARE Classic**. Most phones with an **NXP** chip do, many with a **Broadcom** chip do not. You can check this with the free app “MIFARE Classic Tool”.
- The menu is the same as on the PC: **File → Settings …**.

> **Tip:** If reading is unreliable, try different places on the back of the phone. The NFC antenna is in a different place on every model, often at the top center or around the camera.

10 Updates

At startup the app asks GitHub in the background whether there is a newer version. Only the **number of the latest version** is requested; nothing else is transmitted. You can switch the check off on the "General" tab and trigger it manually at any time with **Check for updates now**.

If there is a new version, a dialog **"New version available"** opens:

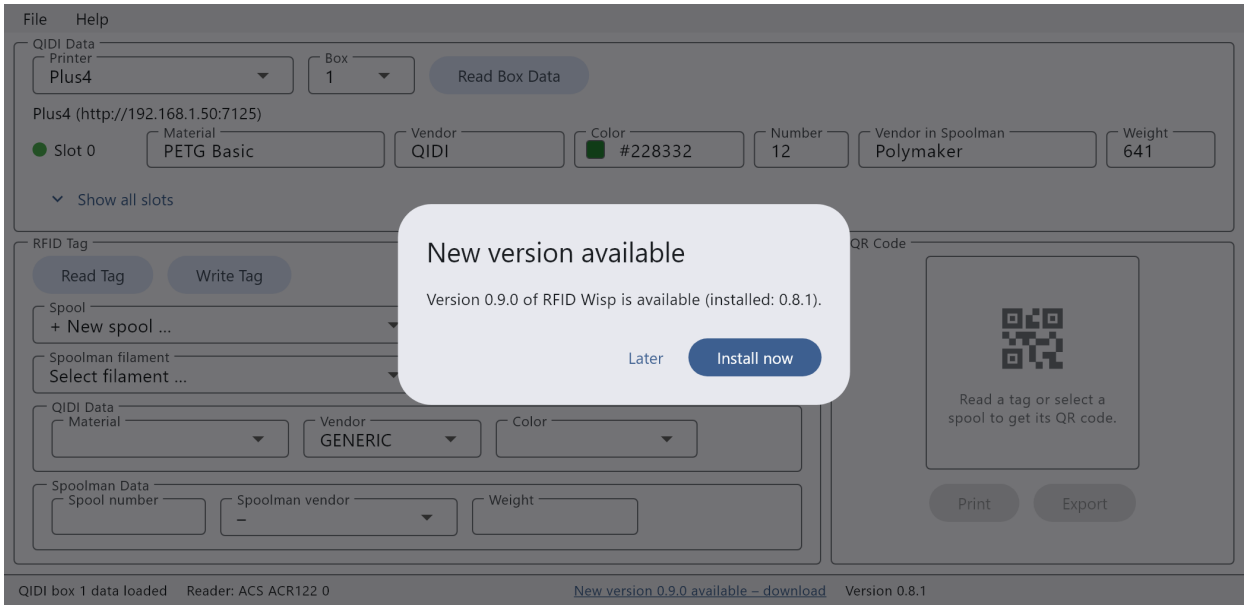


Figure 19: "New version available" dialog with "Later" and "Install now"

- **Install now**: The app downloads the version ("Downloading …"), verifies the **checksum** and starts the installation ("Starting the installation …"). On Windows this uses the installer, on Linux the `.deb` package and on Android the APK. On Android you have to allow the app to install apps beforehand.
- **Later**: The dialog closes. The status bar keeps the link "New version … available – download", which leads to the download page.

The **portable** versions (Windows ZIP, Linux tar.gz) cannot update themselves. The dialog shows the hint "This installation cannot update itself. Download the new version from the download page." and the **Open download page** button.

Error message	Meaning
The download failed.	A network problem. Try again later.
The downloaded file is damaged or does not match the published one.	The checksum does not match; the file is not installed.
Allow RFID Wisp to install apps in the Android settings, then start the installation again.	Android only: the "Install unknown apps" permission for RFID Wisp is missing.
The installation could not be started.	A general error; install by hand from the download page.

11 Optional: the Klipper Module rfid_bridge

`rfid_bridge.py` is an **optional add-on module** that runs **on the printer** in Klipper, not on the PC. You only need it if you want

- the "QIDI Data" frame to show **Number** and **Vendor in Spoolman** per slot (on the **QIDI Q2** the module is the only source of spool numbers), and/or
- the **active spool** to be reported automatically to Moonraker/Spoolman, so that it follows the spool changes of the box in Fluidd and Spoolman.

> **Without Spoolman:** The automatic reporting of the active spool requires the `[spoolman]` component to be set up in `moonraker.conf`. Without Spoolman the module cannot be used for reporting; showing "Number" in "QIDI Data" still works.

11.1 Installation

1. Download `rfid_bridge.py` from the release page (it is attached to every version).
2. Copy the file on the printer into Klipper's `klippy/extras/` directory.
3. Add to `printer.cfg`:

```
[rfid_bridge]
box_stepper_count: 4
moonraker_url: http://127.0.0.1:7125    # optional, default value
moonraker_api_key:                      # optional, only if Moonraker
                                        # requires login even locally
```

4. Restart the Klipper **service**: `sudo systemctl restart klipper`. A Klipper `RESTART` does **not** reload the module. This also applies after every replacement of the file.
5. Check in `moonraker.conf` that `[spoolman]` is set up (for reporting the active spool).

With the command `RFID_BRIDGE_STATUS` in the Klipper console (or `GET /printer/objects/query?rfid_bridge` through Moonraker) you can see the captured state and the **version of the module**. The module has a version number of its own, independent of the version number of the app.

The table shows which version of `rfid_bridge.py` is attached to which version of the app. The file on the printer is not updated with the app: the version installed is the one you copied last (step 2).

App version	Module version
0.8.8	1.1.8
0.8.7	1.1.8
0.8.6	1.1.8
0.8.5	1.1.8
0.8.4	1.1.7
0.8.3	1.1.4
0.8.2	1.1.4

App version	Module version
0.8.1	1.1.3
0.8.0	1.1.2
0.7.0	1.0.3
0.5.2 to 0.6.1	1.0.0 to 1.0.2

11.2 How it works

- The module **listens in** on what QIDI's firmware already reads from the RFID chips of the slots. It changes nothing, modifies no QIDI file and is gone again when the Klipper service restarts (it stays active after a Klipper `RESTART`, though).
- From the 16 bytes it takes the **spool number** and reports it to Moonraker (`/server/spoolman/spool_id`). An empty tag is reported as “no spool”; the active spool is then cleared.
- It reports the spool of the currently **active slot**: when another slot becomes active (tool change), when a new reading arrives for the active slot and when a **print starts**. The report runs in the background, a slow Moonraker does not block Klipper.
- It remembers whether a slot is **occupied** (filament sensor at the slot). If a spool is removed, the cached tag is discarded and the active spool is cleared. A reading for a slot that is reported as empty is ignored.

Tip: Enable **“Check upon startup”** in the settings of the **QIDI Box** (“Filament” page, gear icon in Fluidd or on the printer's touchscreen). The box then re-reads every spool each time the printer restarts (takes about 1–2 minutes) and the active spool is reported reliably. With **“Re read filament information”** you force a single slot to be read again, as long as its filament is not being fed through the box.

Note: After a start, a slot can stay without a spool number even though a spool is inserted. While reading the slots, the box occasionally returns the tag of the neighbouring slot instead of the tag of the slot it is about to activate (seen on a Plus4: slot 1 returned the tag of slot 0, after the Klipper service restart and after a cold start). The display and Fluidd then show the neighbour's data for this slot, and `rfid_bridge` has no read of its own for it. It then reports no spool number for this slot (it does not assign the neighbour's number). Unload the spool from this slot and load it again, or press “Re read filament information” for the slot (only while its filament is not being fed through the box) or re-read the slot in the app with the symbol next to the slot (section 5.2). Afterwards the display, Fluidd and the app show the right spool. The cause lies in the box firmware, not in the bridge; the next note explains it.

Note: Why this happens after a start: two slots share one RFID reader, slots 0 and 1 the first, slots 2 and 3 the second. The reader reads the tag that happens to be in front of it and does not know which spool it belongs to. The box reads the slots one after the other. When slot 0 is done, its spool stands still, and its tag can still be in front of the reader when the box starts with slot 1. The box firmware starts reading before it moves the spool of slot 1, reads the tag of the neighbouring spool twice in a row and stores it for slot 1. Whether this happens depends on where the tag of the neighbouring spool happens to be, that is on how full the spool is and how it was put on. That is why it shows up on one printer again and again and never on another. If you re-read the slot on its own later, the tag of the neighbouring spool is usually no longer in front of the reader, and the slot is read correctly. In tests on two Plus4 printers the error occurred on one of them at eight cold starts in a row and on the other at one of three cold starts with the same set-up of spools. Afterwards it stayed away once the spools had been taken out and put back in, turned a little, or the tag of the spool in slot 0 had been stuck on in another place. So what matters is how the neighboring spool lies in its holder, not a fixed place on the spool. If a slot regularly stays without a number, take the spool of the neighboring slot on the same reader (slot 0 for slot 1, slot 2 for slot 3) out, put it back in, or turn it a little before the start; that may help. This was only tested for slots 0 and 1. This is our interpretation of the firmware's behavior; QIDI has not confirmed it.

Warning: Do not add an extra `SET_ACTIVE_SPOOL` macro call in the slicer. It runs later in the print than the report of `rfid_bridge` and overwrites the spool recognized from the tag with the fixed number from the filament profile. The same applies to the macro `CLEAR_ACTIVE_SPOOL`: it sets the active spool to “none” and would delete the spool reported by `rfid_bridge` again, for example in the start G-code or in a filament change macro. Moonraker provides both macros with the `[spoolman]` component (`SET_ACTIVE_SPOOL ID=<spool number>` and `CLEAR_ACTIVE_SPOOL`). They are described in the Moonraker documentation: https://moonraker.readthedocs.io/en/latest/configuration/#spoolman

Tip: Pass the first slot in the slicer's start G-code. Otherwise QIDI's firmware uses slot 0 (default `EXTRUDER=0`), loads it first and the spool of slot 0 is booked, even if the first colour of the print is on another slot. To fix this, add `EXTRUDER=[initial_no_support_extruder]` to the end of the `PRINT_START` call in the printer's start G-code (Orca-based slicers such as QIDI Studio or OrcaSlicer replace the placeholder with the number of the first slot). Some profiles call `BOX_PRINT_START` directly and already pass the value (for example the Max4 profile); nothing needs to be added there.

Note: While the printer heats, Fluidd or Spoolman may still show the old spool. The box activates the first slot of a print only after hotend and bed have heated up, and only then does `rfid_bridge` learn which spool the print uses. This is harmless, because booking only happens once filament is extruded. If the box does not activate any slot, the module reports the previously active spool 2 minutes after hotend and bed have reached their target (10 minutes after the print start at the latest).

11.3 Optional settings

Option	Default	Meaning
`runout_present_value`	`0`	Value of the `runout_button` sensor that means “filament present”. `0` is confirmed on the Q2, Plus4 and Max4.

Option	Default	Meaning
`ignore_reads_on_empty_slot`	`True`	Ignores readings that are assigned to a slot the sensor reports as empty.
`occupancy_poll_interval`	`0.5`	How often (in seconds) the sensors are polled.

12 Troubleshooting

12.1 Problems and solutions

Problem / message	Possible cause and solution
“Read Tag” and “Write Tag” are gray	No default reader chosen: **File → Settings → Reader**. Or an action is still running (busy pointer).
Only “Write Tag” is gray for a moment	After reading, the app is still asking Spoolman for the spool of the tag (section 6.3). Wait a moment.
“Default reader not found”	Device not connected or driver missing. Connect it and use **Search again** in the settings.
“Smart card service not available”	Windows: start the “Smart Card” service. Linux: install and start `pcscd` (section 2.3).
The reader does not appear on Linux (ACR122U)	Blacklist the kernel driver `pn533_usb`, add a polkit rule for `pcscd` (section 2.3). Check with `pcsc_scan`.
The reader does not appear in the list although connected	The app only lists devices that can really read and write RFID tags. Check the driver; use **Search again**.
“No RFID tag detected on the reader.”	Hold the tag closer to the reader or put it on again. Put on only **one** tag.
“The tag … is not a MIFARE Classic 1K”	Wrong tag type. Only MIFARE Classic 1K tags are supported (not Ultralight, NTAG or similar).
“Cannot authenticate to sector …”	The tag is secured with an unknown key. Use another tag.
“Reading / Writing the tag failed.”	Place the tag on the center of the reader and try again. If it keeps failing, use another tag.
Error dialog when reading: tag “probably not written by this application”	The tag contains values that appear in no known filament table. Write it again (it is overwritten).
Red message “… – not connected” in the “QIDI Data” frame	Check the Moonraker address and the network, switch on the printer, then **Check the printer again**.
Red message “Box switched off on the printer …” below the printer in the “QIDI Data” frame	The printer reports its box as switched off (`enable_box = 0`), often after a print from the external spool. Switch the box on again on the printer, then **Check the printer again**.
“Could not read the QIDI box”	The box does not answer. Check whether the printer recognizes the box (Fluidd) and read again.
“The printer has no connection to Spoolman”	Moonraker on the printer cannot reach Spoolman. Check the network, the Spoolman server and `[spoolman]` in `moonraker.conf` (section 5.3).
“Number” and “Vendor in Spoolman” empty in the “QIDI Data” frame	`rfid_bridge` is not running on the printer, was not loaded by a service restart after copying, or the tag has no number.
“Weight” empty in the “QIDI Data” frame	No Spoolman address, “Use Spoolman” switched off, slot without a number, or Spoolman does not know the spool.
“Could not load data from Spoolman: …”	Check the Spoolman address, the server and the network (“Spoolman” tab).

Problem / message	Possible cause and solution
“Spoolman sync skipped: …”	The spool could not be created in Spoolman when writing; the tag was written with the number in the form (section 6.7).
“No matching filament found in Spoolman for this …”	Create the filament in Spoolman or directly choose a **Spoolman filament**.
A new spool from Spoolman is missing from the list	Press **Read Tag** once or restart the app so that the list is reloaded.
The form suddenly became empty	The tag that was read was taken off (Windows/Linux), the Spoolman setting was changed, or **+ New spool …** was chosen.
Material or color do not match the Spoolman filament	The QIDI lists are limited. The app picks the closest QIDI material or color; adjust the selection by hand.
QR Code frame shows no code	Material, color or spool number is missing (section 7.1). For a new spool with Spoolman, only after writing.
“Print” is gray	No printer set up in the system. Save as a PDF with **Export**.
Android: “NFC is switched off”	Turn on NFC in the device settings.
Android: tag is not detected	The phone's NFC chip may not support MIFARE Classic (section 9.1), or you are holding the tag in the wrong place.
Windows blocks the start of the app	See section 2.2 (SmartScreen / Smart App Control).

12.2 The spool number is above 999

The tag only has room for numbers from **0 to 999**. If Spoolman assigns a higher spool number (because you have already created a very large number of spools there), it cannot be written to the tag, and the app reports an error. The only workaround is to work without Spoolman or to manage the spools in Spoolman so that the numbers stay below 1000.

12.3 The log (Windows only)

On Windows the app writes every status message with a timestamp to the file `%APPDATA%\RFIDwisp\status.log`. **Help → Open log** opens it in the editor. This helps with troubleshooting, because messages are otherwise overwritten by the next one. When the file reaches 10 MB, it is renamed to `status_YYYY-MM-DD_HH-mm-ss.log` and a new one is started.

Appendix A Layout of the Tag Data

For information: RFID Wisp writes **sector 1** of the MIFARE Classic 1K tag with a block of **16 bytes** that the QIDI box reads.

Byte	Contents	Remark
0	Material	Code of the material in the printer's filament table
1	Color	Code from the QIDI color palette
2	QIDI vendor	Code of the vendor (0 = “GENERIC”)
3–10	–	unused, 0
11–12	Last weight	in whole grams, 16 bit, big-endian (new spool: initial weight; existing spool: remaining weight). 0 without Spoolman.
13	Spoolman vendor	Spoolman ID of the vendor (0 to 255), 0 = none
14–15	Spool number	0 to 999, 16 bit, big-endian

A tag whose 16 bytes are all zeros counts as **blank**. If material, color or vendor match no known table, the tag counts as “not written by this application”.

Appendix B Status Bar Messages

Message	When
Ready	After startup, as long as nothing has happened yet.
QIDI box <n> data loaded	A box was read successfully.
RFID tag read (spool #<n>)	A tag was read.
RFID tag written (spool #<n>)	A tag was written and read back for verification.
RFID tag removed, fields cleared	The tag that was read was taken off the reader (Windows/Linux).
Blank RFID tag detected. Ready to write new spool data.	A blank tag was read.
Hold the RFID tag to the back of the device …	Android only: hold the tag to the phone now.
QR code of spool #<n> saved: <path>	PDF export succeeded.
QR code of spool #<n> sent to printer <name>	The label was sent to the printer.
Spoolman sync skipped: <reason>	The spool could not be created in Spoolman when writing; the tag was written with the number in the form.
Could not load data from Spoolman: <reason>	Spoolman was not reachable when loading.
Error fetching vendors from Spoolman	In the “QIDI Data” frame the vendor names could not be loaded.
The printer has no connection to Spoolman	Moonraker on the printer cannot reach Spoolman.
QIDI box data not available: <reason>	The printer is not reachable.
Could not read the QIDI box: <reason>	Reading a box failed.
Re-reading slot <n> …	The box reads the tag of the slot once more; this takes about half a minute.
Slot <n> re-read	The box has read the slot; the app then reads the box again.
Slot <n> could not be re-read: <reason>	The printer is printing, the box is already reading, the box refuses the command (for example because filament is in the extruder) or the printer is not reachable.

Appendix C Glossary

Term	Explanation
APK	Installation file for Android.
CCID / PC/SC	Standards through which a computer talks to smart card and RFID readers.
Klipper	The printer's firmware software that controls the print.
MIFARE Classic 1K	A widely used RFID chip with 1 KB of memory in 16 sectors. The QIDI box uses it.
Moonraker	Web interface to Klipper; the basis of Fluidd and Mainsail.
NFC	Near-field technology with which phones read RFID tags.
Sector	Memory area of a MIFARE tag. RFID Wisp uses sector 1.
Slot	Position of a spool in the QIDI Box.
Spoolman	Server for managing filament spools.
Tag	The RFID chip on the spool.